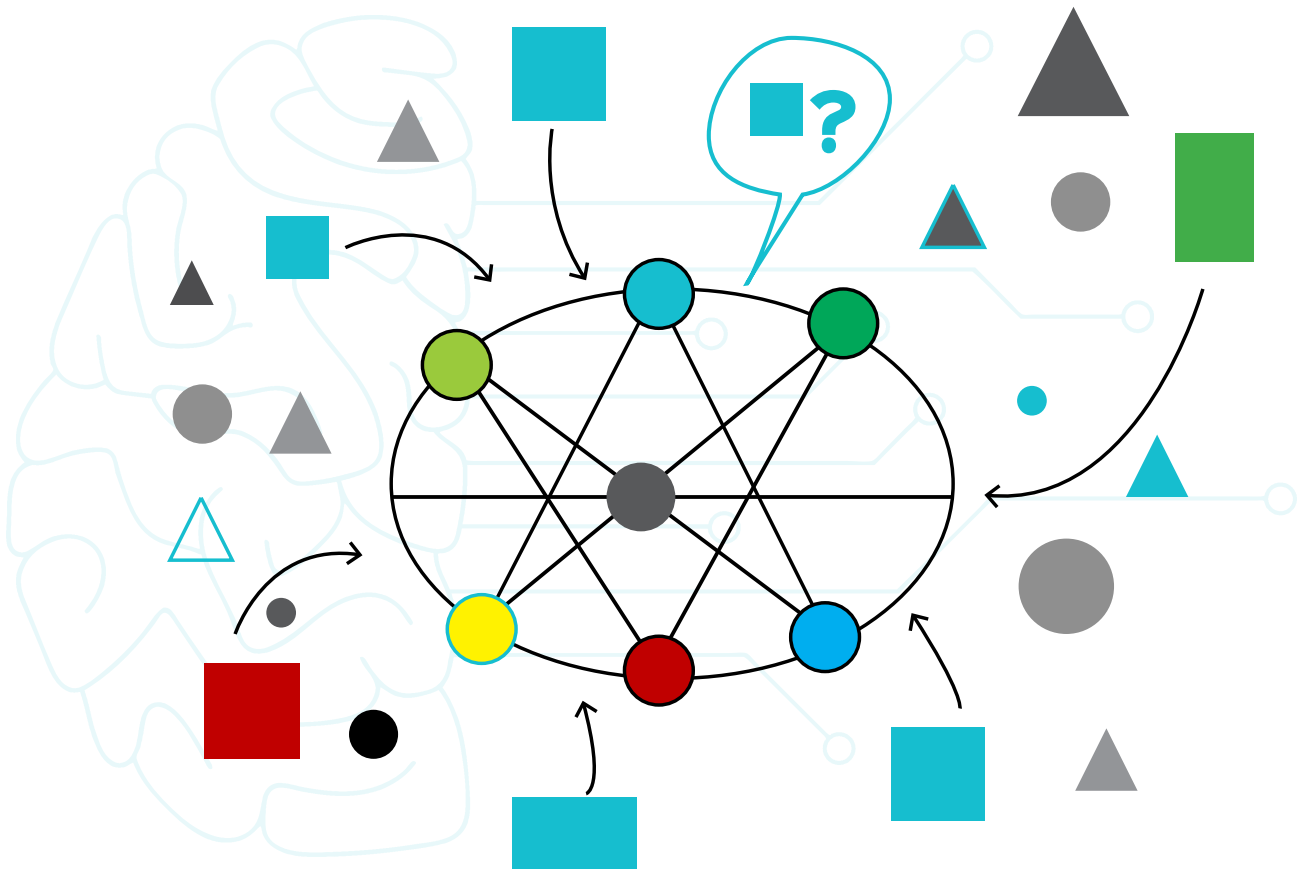


What is Federated Learning?

Federated learning is a distributed machine learning approach for developing and training models on a global scale. This article introduces a few open source frameworks that help to explore it, which every newbie must know of.



Federated learning helps to develop and train models for distributed machine learning. Loosely translated, federation means a group of objects working towards the same common goal. In federated learning, the model is developed with the help of updates from a large group of devices, usually phones. Prior to the advent of federated learning, devices leveraging various artificial intelligence solutions sent repeated messages embedded with the data

stored on the device. This led to two major bottlenecks:

- *Communication:* Not all devices in question had constant Internet access. Moreover, the bandwidth was very often insufficient for such a transfer.
- *Privacy:* Any third party actor with a malicious intent could figure out a way to access the data coming in from a consumer.

To address all of this and much more, Google introduced federated

learning. Originally built into the Google keyboard, it quickly piqued the interest of the research community. The benefits of federated learning are:

- Leverages the on-device compute power of modern-day electronics
- Protects the privacy of the customer with zero data transfers
- Can function in a decentralised architecture to handle single-point failures
- Trains over a high number of clients even with slow or low-quality Internet